

MediAssist AI Enterprise Architecture Breakdown Document

ARCHITECTURE OVERVIEW

MediAssist AI is an enterprise healthcare copilot built using RAG, Multi-Agent AI, Multimodal AI, MCP integration, and enterprise retrieval systems.

COMPONENTS

1. User

- **Role:** Represents healthcare stakeholders interacting with the platform for operational and analytical tasks.
- **Input:**
 - Natural language queries
 - Medical reports
 - Prescription images
 - Lab reports
 - Operational requests
- **Output:** Natural language queries, uploaded reports, prescriptions, lab reports, and user requests.

2. Streamlit UI Frontend

- **Role:** Provides a user-friendly interface for healthcare staff to interact with the AI system, submit queries, and upload documents/images.

Responsibilities

- Chat interface
- File uploads
- Response visualization
- Agent execution display
- Evaluation dashboard

- **Input:**
From Users:
 - Text queries
 - PDF documents
 - Images
- **Output:** Structured API requests containing user query, uploaded files, session information, and metadata.

3. FastAPI Backend Gateway

- **Role:** Handles API requests, validates inputs, manages sessions, and routes requests to the orchestration layer.

Responsibilities:

- Receive requests from UI
 - Validate requests
 - Route requests
 - Manage sessions
 - Forward requests to orchestration layer
- **Input:** UI requests.
 - User query
 - Uploaded files
 - Session metadata
- **Output:** Validated and standardized request payloads forwarded to the orchestration layer.

4. LangGraph Supervisor

- **Role:** Orchestrates workflow execution and coordinates communication between multiple AI agents.

Responsibilities

- Workflow management
 - Agent coordination
 - Agent selection
 - State management
- **Input:** User request.
- **Output:** Workflow execution plan, agent routing instructions, and state management context.

5. Planner Agent

- **Role:** Analyzes user intent and creates an execution plan by selecting the required agents and tools..

Responsibilities

- Intent understanding
- Task decomposition
- Workflow planning
- **Input:** User query.
- **Output:** Task breakdown, execution strategy, selected agents, and workflow sequence.

6. Retriever Agent

- **Role:** Retrieves relevant information from vector databases using semantic search and RAG techniques.

Responsibilities

- Semantic search
- Vector retrieval
- Context grounding
- Document ranking
- **Input:** Search query.
- **Output:** Relevant document chunks, SOPs, policies, contextual knowledge, citations, and retrieved enterprise content.

7. MCP Agent

- **Role:** Accesses structured healthcare data through MCP tools and enterprise database integrations.

Responsibilities

- Tool execution
- SQL abstraction
- Enterprise integration
- **Input:** MCP tool request.
- **Output:** Structured healthcare records such as patient history, appointments, billing details, prescriptions, and lab results.

8. Vision Agent

- **Role:** Processes medical images, prescriptions, and reports to extract meaningful healthcare information.

Responsibilities:

- OCR
 - Prescription extraction
 - Medical report understanding
 - Information extraction
- **Input:** Prescription and report images.
Images
 - Prescription image
 - Lab report image
 - Medical report image

- **Output:** Extracted text, medicines, diagnoses, observations, report summaries, and structured image insights.

9. Vector Database

FAISS / ChromaDB

- **Role:** Stores document embeddings and enables semantic retrieval of enterprise knowledge.
- **Input:** Embeddings generated during ingestion
- **Output:** Ranked and semantically relevant document chunks based on similarity search.

10. PostgreSQL Database

- **Role:** Stores structured healthcare data including patients, appointments, billing, and prescriptions.
- **Input:** MCP queries.
- **Output:** Patient records, appointment data, billing information, insurance details, admission history, and analytics data.

11. MCP Server

- **Role:** Provides secure and standardized access to enterprise systems through tool-based interactions.

Responsibilities

- Tool registration
- Tool execution
- Database communication
- Access control
- **Input:** Tool invocation request
- **Output:** Tool execution results, database query responses, and standardized JSON outputs.

12. LLM Processing Layer

- **Role:** Performs language understanding, reasoning, summarization, and response generation.

Responsibilities:

- Understanding queries
- Tool calling
- Reasoning support
- Agent collaboration
- **Input:**
 - Retrieved documents
 - Structured records
 - Vision outputs
 - User prompts
- **Output:** Intent understanding, reasoning context, summaries, generated insights, and intermediate AI responses.

13. Reasoning Agent

- **Role:** Combines information from multiple sources to generate contextual and grounded responses.
- **Input:** Outputs from Retriever, MCP, and Vision agents.
- **Output:** Consolidated context, recommendations, healthcare insights, clinical summaries, and response drafts.
 - Clinical summary
 - Recommendations
 - Final response draft

14. Evaluator Agent

- **Role:** Validates response quality, detects hallucinations, and ensures accuracy and relevance.

Responsibilities

- Hallucination detection
 - Grounding validation
 - Faithfulness checking
 - Relevance evaluation
- **Input:** Generated answer, Retrieved evidence
 - **Output:** Faithfulness score, relevance score, grounding score, hallucination assessment, and quality validation results.

15. Observability Layer

- **Role:** Monitors system performance, logs activities, and tracks agent execution flows.

Responsibilities

- Logging
 - Monitoring
 - Agent tracing
 - Workflow debugging
- **Input:** Agent execution events
 - **Output:** Execution logs, traces, performance metrics, error reports, latency statistics, and monitoring dashboards.

16. Response Agent

- **Role:** Formats and delivers the final validated response to the user.

Responsibilities:

- Applying guardrails
- Formatting responses
- Security validation
- Output generation

- **Input:** Validated response from Evaluator Agent
- **Output:** Final formatted response, recommendations, summaries, citations, and user-ready output.